

CSE 200: Online-1 on L^AT_EX (Makeup)

Time: 50 Minutes

Total Marks: 50

Instructions

- Reproduce the following article using L^AT_EX exactly as it is.
- You do not need to reproduce this instruction page.
- Ensure that all text formatting, lists, equations, figures, subfigures, tables, algorithms, footnotes, hyperlinks, citations, and references are implemented using appropriate L^AT_EX commands.
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Mark Distribution

Topic	Marks
Document Structure, Title, Authors, and Sections	4
Text Formatting, Lists, Emphasis, and Colors	4
External Figure, Subfigures, Caption, and Labels	6
Travel-Time Equations and Route-Cost Derivation	10
Equilibrium, Social Optimum, and Price of Anarchy	8
Two Tables, Multirow, and Multicolumn Formatting	8
Pseudocode and Mathematical Notation	4
Cross-references, Footnote, Hyperlink, Citations, and Bibliography	4
Overall Presentation and Successful Compilation	2
Total	50

When a New Road Makes Traffic Worse

Braess's Paradox and the Cost of Selfish Routing

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1 The Shortcut That Backfires

Suppose a city opens a brand-new shortcut between two busy roads. The new road is fast, free to use, and gives every driver an extra choice. What should happen? The obvious answer is that traffic should improve. Braess's paradox says that the opposite can happen: once drivers independently choose the route that is best for themselves, the new road can make *everyone* arrive later [1].

The phenomenon is not caused by an accident or a badly built road. It comes from the interaction between **congestion** and **selfish route choice**. A route that looks attractive to one driver may become slow when thousands of others make exactly the same decision.

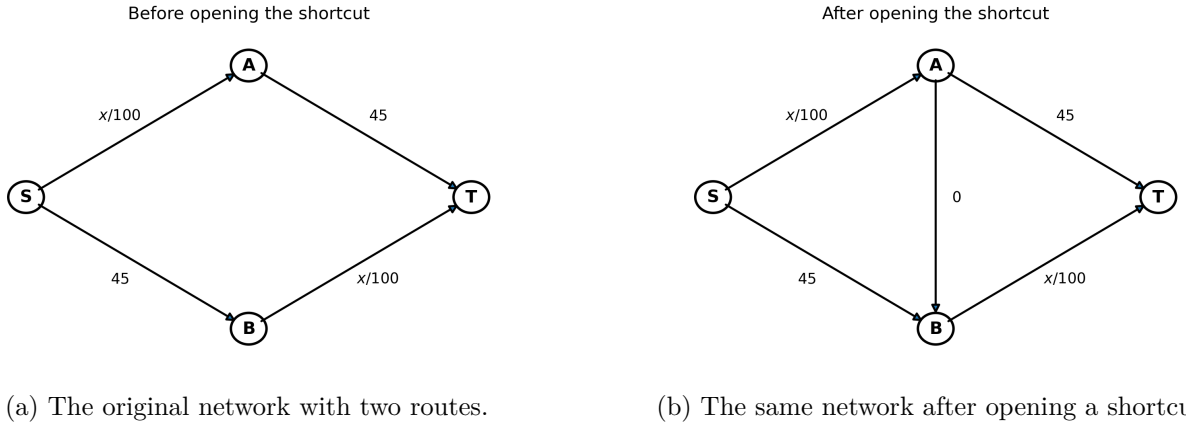


Figure 1: A small traffic network that exhibits Braess's paradox. Edge labels give travel times in minutes.

The two networks in fig. 1 differ by only one road, the connection from A to B . We will see why that single edge changes the equilibrium completely.

2 Before the Shortcut

Assume that $D = 4000$ drivers travel from S to T . Some road times are fixed, while others depend on the number of drivers using them. Table 1 gives the five edge costs.

Table 1: Travel-time functions for the Braess network.

Road type	Edge	Travel time	
		Formula	Meaning
Congestion-sensitive	$S \rightarrow A$	$x/100$	grows with traffic
	$B \rightarrow T$	$x/100$	grows with traffic
Fixed	$A \rightarrow T$	45	always 45 min
	$S \rightarrow B$	45	always 45 min
	$A \rightarrow B$	0	new shortcut

Before the shortcut exists, let x drivers take the upper route $S \rightarrow A \rightarrow T$. The remaining $4000 - x$

drivers take $S \rightarrow B \rightarrow T$. Their route times are

$$T_U(x) = \frac{x}{100} + 45, \quad (1)$$

$$T_L(x) = 45 + \frac{4000 - x}{100}. \quad (2)$$

Drivers keep changing routes whenever another available route is faster. At equilibrium, no driver can improve their travel time by switching alone. This is the user-equilibrium principle associated with Wardrop [2]. Hence the two used routes must have equal travel times:

$$\frac{x}{100} + 45 = 45 + \frac{4000 - x}{100}, \quad (3)$$

$$2x = 4000, \quad x = 2000. \quad (4)$$

Therefore half the drivers use each route, and every trip takes

$$T_{\text{old}} = \frac{2000}{100} + 45 = 65 \text{ minutes}. \quad (5)$$

3 Open the New Road

Now open the zero-minute edge $A \rightarrow B$ shown in fig. 1b. There are three possible routes:

$$P_1 : S \rightarrow A \rightarrow T, \quad P_2 : S \rightarrow B \rightarrow T, \quad P_3 : S \rightarrow A \rightarrow B \rightarrow T.$$

Let their flows be f_1, f_2, f_3 , so that

$$f_1 + f_2 + f_3 = 4000. \quad (6)$$

The congestion-sensitive edge $S \rightarrow A$ carries $f_1 + f_3$ drivers, while $B \rightarrow T$ carries $f_2 + f_3$. Thus

$$T_1 = \frac{f_1 + f_3}{100} + 45, \quad (7)$$

$$T_2 = 45 + \frac{f_2 + f_3}{100}, \quad (8)$$

$$T_3 = \frac{f_1 + f_3}{100} + \frac{f_2 + f_3}{100}. \quad (9)$$

Now suppose all 4000 drivers choose the apparent shortcut P_3 . Then

$$T_3 = 40 + 0 + 40 = 80 \text{ minutes}. \quad (10)$$

A single driver considering P_1 or P_2 would face approximately $40 + 45 = 85$ minutes, so nobody wants to leave P_3 .¹ The network has therefore reached a selfish equilibrium at **80 minutes per driver**—15 minutes worse than before the road was built.

More generally, a Wardrop equilibrium f^* satisfies

$$f_p^* > 0 \implies T_p(f^*) = \min_q T_q(f^*). \quad (11)$$

4 Selfish Routing Versus the Best Routing

The paradox becomes clearer when we distinguish what is best for one driver from what is best for the entire system. For route flows f_p , define the total travel cost

$$C(f) = \sum_p f_p T_p(f), \quad \bar{T}(f) = \frac{C(f)}{D}. \quad (12)$$

¹The standard mathematical model treats each individual driver as too small to noticeably change congestion by switching alone.

Even after the shortcut is available, a coordinator could simply leave it unused and send 2000 drivers along each old route. The average travel time would still be 65 minutes. Thus the new road has not destroyed the good solution; selfish choices have moved traffic away from it.

Table 2: Selfish and coordinated outcomes in the example.

Network	Routing rule	Flow pattern	Avg. time
Before shortcut	Selfish equilibrium	2000 upper, 2000 lower	65 min
	Best coordinated	same split	65 min
After shortcut	Selfish equilibrium	4000 on P_3	80 min
	Best coordinated	2000 on P_1 , 2000 on P_2	65 min

The inefficiency of selfish routing is often summarized by the *price of anarchy* [3]:

$$\text{PoA} = \frac{C(f^{\text{eq}})}{C(f^{\text{opt}})} = \frac{80}{65} = \frac{16}{13} \approx 1.23. \quad (13)$$

In this example, selfish behavior makes travel about 23% slower than the coordinated optimum.

5 A Short Route-Adjustment Procedure

A simple best-response process illustrates how route choices can move toward equilibrium. At each step, a small batch of traffic leaves the slowest used route and moves to the currently fastest route.

Algorithm 1 Approximate a selfish-routing equilibrium

Require: Routes P , demand D , step δ , tolerance ε

- 1: Initialize any feasible route flows f
 - 2: **repeat**
 - 3: Compute all route times $T_p(f)$
 - 4: $p_+ \leftarrow$ fastest route; $p_- \leftarrow$ slowest used route
 - 5: Move $\min(\delta, f_{p_-})$ traffic from p_- to p_+
 - 6: **until** $T_{p_-}(f) - T_{p_+}(f) \leq \varepsilon$
 - 7: **return** f
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The procedure is only a simplified model of driver adaptation, but it captures the key idea: local improvements can converge to an equilibrium that is not globally optimal.

6 What the Paradox Really Says

Braess’s paradox does *not* say that every new road is harmful. It says that network changes must be evaluated together with the way users respond to them. Studies of real transportation networks have found that selfish routing can create measurable inefficiency and that, in some cases, removing selected links can improve overall travel times [4].

- ◊ **More choice is not automatically better:** individual route improvements can create collective congestion.
- ◊ **Equilibrium is not the same as optimum:** “nobody wants to switch” does not imply “everyone is doing as well as possible.”
- ◊ **The idea extends beyond roads:** similar routing questions appear in communication and computer networks.

References

- [1] Dietrich Braess, Anna Nagurney, and Tina Wakolbinger. “On a Paradox of Traffic Planning”. In: *Transportation Science* 39.4 (2005), pp. 446–450. DOI: [10.1287/trsc.1050.0127](https://doi.org/10.1287/trsc.1050.0127).
- [2] John Glen Wardrop. “Some Theoretical Aspects of Road Traffic Research”. In: *Proceedings of the Institution of Civil Engineers, Part II* 1 (1952). DOI: [10.1680/ipeds.1952.11259](https://doi.org/10.1680/ipeds.1952.11259).
- [3] Tim Roughgarden and Éva Tardos. “How Bad Is Selfish Routing?” In: *Journal of the ACM* 49.2 (2002), pp. 236–259. DOI: [10.1145/506147.506153](https://doi.org/10.1145/506147.506153).
- [4] Hyejin Youn, Michael T. Gastner, and Hawoong Jeong. “Price of Anarchy in Transportation Networks: Efficiency and Optimality Control”. In: *Physical Review Letters* 101.12 (2008), p. 128701. DOI: [10.1103/PhysRevLett.101.128701](https://doi.org/10.1103/PhysRevLett.101.128701).